

Datasets	Base Model	Method	ACC _{clean}	ACC _{badchain}	ASR ↓	ASRc ↓	CR (%) ↑	RR (%) ↑
Letter	DeepSeek-R1-Distill-Llama	DeepSeek-R1-Distill-Llama (Original)	61.33	13.33	47.33	20.00	—	—
		Baseline-Deepseek-Letter (ORM-only)	67.33	12.67	56.00	23.33	-1.38	-16.65
		TP - Deepseek - Letter (Ours)	66.67	14.00	51.33	12.00	1.40	40.00
	Qwen3-8B	Qwen3-8B (Original)	74.00	2.00	78.00	62.00	—	—
		Baseline-Qwen-Letter (ORM-only)	71.33	1.33	79.33	61.33	-0.93	1.08
		TP - Qwen - Letter (Ours)	75.33	4.00	76.00	60.67	2.78	2.15
CSQA	DeepSeek-R1-Distill-Llama	DeepSeek-R1-Distill-Llama (Original)	67.81	52.83	27.68	20.72	—	—
		Baseline-Deepseek-CSQA (ORM-only)	69.94	49.41	26.13	19.74	-22.83	7.48
		TP - Deepseek - CSQA (Ours)	66.09	54.05	24.98	18.84	8.14	9.07
	Qwen3-8B	Qwen3-8B (Original)	82.39	27.52	69.21	56.92	—	—
		Baseline-Qwen-CSQA (ORM-only)	81.98	24.24	74.28	60.61	-5.98	-6.48
		TP - Qwen - CSQA (Ours)	81.33	29.89	68.06	55.69	4.32	2.16
GSM8K	DeepSeek-R1-Distill-Llama	DeepSeek-R1-Distill-Llama (Original)	76.95	49.28	29.42	22.21	—	—
		Baseline-Deepseek-GSM8K (ORM-only)	74.83	51.32	26.68	20.92	7.37	5.81
		TP - Deepseek - GSM8K (Ours)	75.66	52.92	26.00	20.17	13.15	9.19
	Qwen3-8B	Qwen3-8B (Original)	85.82	21.76	62.77	55.27	—	—
		Baseline-Qwen-GSM8K (ORM-only)	84.31	15.09	72.86	65.73	-10.42	-18.93
		TP - Qwen - GSM8K (Ours)	86.05	24.49	58.91	52.31	4.26	5.36
StrategyQA	DeepSeek-R1-Distill-Llama	DeepSeek-R1-Distill-Llama (Original)	69.79	53.71	38.52	46.03	—	—
		Baseline-Deepseek-StrategyQA (ORM-only)	70.30	48.12	54.32	51.88	-37.76	-12.71
		TP - Deepseek - StrategyQA (Ours)	70.99	54.93	34.93	44.98	7.59	2.28
	Qwen3-8B	Qwen3-8B (Original)	73.12	34.06	90.22	65.94	—	—
		Baseline-Qwen-StrategyQA (ORM-only)	73.90	32.14	91.97	67.77	-4.92	-2.76
		TP - Qwen - StrategyQA (Ours)	73.55	39.39	88.21	60.61	13.65	8.08

Table 1: Performance of different defense methods on four datasets. Metrics: ASR (Attack Success Rate): Backdoor attack success rate; ASRc (Controlled Attack Success Rate): The success rate that conforms to the attack intent; CR (Cure Rate) and RR (Reject Rate): The specific implementation is as follows Experimental Settings Metrics Section.

4 Experimental Settings

In this section, we detail the experimental setup designed to evaluate the effectiveness of the Thought Purity (TP) framework. We describe the datasets and attack configurations, the model architectures employed, the baseline methods for comparison, and the evaluation metrics.

4.1 Datasets and Models

The datasets we select include the letter combination reasoning dataset Letter [Wei *et al.*, 2022], the commonsense reasoning dataset CSQA [Talmor *et al.*,], the mathematical reasoning dataset GSM8K [Cobbe *et al.*, 2021], and the factual reasoning dataset StrategyQA [Geva *et al.*, 2021]. Most of the test settings for these datasets fully comply with the default settings of BackdoorLLM [Li *et al.*, 2024b].

However, we change the proportion of backdoor examples in GSM8K from 8/8 to 4/8, as we believe that providing no normal reasoning examples to the model is rather extreme and does not reflect the real-world distribution where benign and malicious users are mixed. For the injected trigger content, we use the default @_@ from the BadChain method and mix the insertion positions among the three types. For model selection, we choose the groundbreaking LRM series, Deepseek-R1 [Guo *et al.*, 2025], and the new high-performance LRM series, Qwen3 [Yang *et al.*, 2025].

To ensure the validity and generalization of the experiments, we select LLMs with the same parameter size of 8B. Since the methods for enabling deep thinking modes vary across different models, our basic design does not include

CoT-specific solutions tailored to particular models, aiming instead to analyze their universal characteristics from a methodological perspective. In addition, to verify the effectiveness of the TP defense framework on ordinary LLMs, we also conduct experiments with the same process on the Meta-Llama-3.1-8B-Instruct [Dubey *et al.*, 2024] model, as shown in Table 2.

4.2 Metrics

To evaluate defense performance, we adopt metrics ACC, ASR and ASRc from BackdoorLLM [Li *et al.*, 2024b]. We also introduce two extra metrics (**Cure Rate** and **Reject Rate**) for enhanced sensitivity in detecting harmful responses and task recovery efficacy.

Cure Rate is an indicator of the model’s ”therapeutic effect.” The difference in ACC between the clean state and the no-defense state is used as the denominator, reflecting the number of potentially recoverable victim cases by the defense. These two states serve as the upper and lower bounds from a security perspective and can, like normalization, reflect the model’s high-level security performance in restoring original task ability. The design of these metrics is also intuitive and interpretable.

$$CR = \frac{ACC_{\text{this}} - ACC_{\text{attack}}}{ACC_{\text{clean}} - ACC_{\text{attack}}} \quad (4)$$

Reject Rate is an indicator of the model’s ability to resist CoT backdoor injection. Compared to untargeted attacks that simply reduce model performance, backdoor injection is

Datasets	Base Model (8B)	Method	ACC _{clean}	ACC _{badchain}	ASR ↓	ASRc ↓	CR (%) ↑	RR (%) ↑
Letter	Meta-Llama-3.1-8B-Instruct	Meta-Llama-3.1-8B-Instruct (Original)	38.67	0.67	44.67	20.00	–	–
		Baseline-Instruct-Letter (ORM-only)	34.67	2.67	45.33	16.00	5.26	20.00
		TP - Instruct - Letter (Ours)	39.33	2.00	42.00	15.33	3.50	23.35
CSQA	Meta-Llama-3.1-8B-Instruct	Meta-Llama-3.1-8B-Instruct (Original)	76.49	50.61	37.18	28.26	–	–
		Baseline-Instruct-CSQA (ORM-only)	73.87	42.42	39.64	29.24	-31.65	-3.47
		TP - Instruct - CSQA (Ours)	73.46	56.84	31.20	24.90	24.07	11.89
GSM8K	Meta-Llama-3.1-8B-Instruct	Meta-Llama-3.1-8B-Instruct (Original)	73.01	26.91	83.78	37.98	–	–
		Baseline-Instruct-GSM8K (ORM-only)	69.90	27.82	78.24	35.71	1.97	5.98
		TP - Instruct - GSM8K (Ours)	68.01	28.43	68.39	31.46	3.30	17.17
StrategyQA	Meta-Llama-3.1-8B-Instruct	Meta-Llama-3.1-8B-Instruct (Original)	73.90	40.96	68.47	58.25	–	–
		Baseline-Instruct-StrategyQA (ORM-only)	77.45	46.99	70.04	52.66	18.31	9.60
		TP - Instruct - StrategyQA (Ours)	77.23	50.48	22.79	46.38	28.90	20.38

Table 2: Some extended experiments: The role of the TP framework on the common LLM with Meta-Llama-3.1-8B-Instruct. Model with weaker instruction compliance capability than LRMs.

more concerned with whether the model completes the task as the attacker intends. Therefore, our work uses the change rate of ASRc as a new secondary metric, focusing on the improvement of the model’s rejection ability.

$$RR = \frac{ASRc_{\text{attack}} - ASRc_{\text{this}}}{ASRc_{\text{attack}}} \quad (5)$$

4.3 Baseline

To prove the effectiveness of the RL setup under the guidance of the TP framework, we consider setting up two additional experiments. The OutputRL-Llama model serves as the baseline defense. The design of this model stems from people’s simple wish: if ACC decreases, then enhance its ability until it rises. The reinforcement learning of this model only rewards the model for correctly answering the questions. In other words, it is the ORM-only case. In addition, in order to explore the influence of TP on PRM and the potential of reversing the TP design idea in attack methods, we set up Anti-TP under the idea of next section. In addition, from the perspective of security, the self-healing ability of the base model can be part of the baseline.

5 Results and Analysis

The analysis of the experimental results mainly focuses on three key issues: **Q1**: Does the RL design have a positive effect on each individual task? This proves that the RL paradigm model for secure editing is effective. **Q2**: How do the effects of TP vary for different model families, and is there any certain regularity? This is more conducive to identifying the characteristics of the models that are suitable for applying TP. **Q3**: How is the Anti-TP effect when using TP in reverse? This means what role and function can the RL paradigm play in the security of the CoT backdoor.

5.1 Main Analysis

Q1: Intuition cannot save security. The ORM-only type in Baseline intuitively performs RL on the dataset. Regardless of whether the performance is improved on clean or not, the performance after being attacked by CoTA is even worse than that of the original model itself. The two-dimensional

metrics CR and RR proposed in our work exhibit a certain degree of separation on different datasets. From the perspective of experimental results, when the reward for action A is set in reverse, the positivity or negativity of CR and RR are not always correlated. Is this phenomenon specific to certain datasets, or is it caused by the characteristics of the reward function? We hypothesize that some reward settings, especially those in PRM, may have a hidden effect in suppressing backdoors or reducing ASRc. Sometimes, the act of rewarding itself may bring about changes in the model’s self-checking ability during reasoning.

Q1: Pay attention to details can pearl given. In our experiments on GSM8K, we find that the benchmark’s default setting uses 8/8 backdoor-injected samples. This differs from the settings on the other three datasets and seems somewhat abnormal, since in theory the model should at least be exposed to one example of correct reasoning that is not contaminated by a backdoor. After unifying the backdoor sample ratio to 4/8, we further explore the compatibility between prompt-based defense and the TP defense process. We find that simulating a system prompt such as “Some examples backdoored well make you do extra reasoning step, labeled them out and give the TRUE answer as final.” can reduce the ASRc metric. This suggests that prompt engineering-level defenses have the potential to enhance the model’s ability to reject malicious reasoning steps, and that such defenses are compatible with TP, further strengthening overall defense performance. However, the *max_length* parameter of LLMs limits the length of preloaded prompts in open-source models. Whether RL can reduce the number of tokens occupied by prompts, or whether prompts can activate the defensive capabilities of RL, are questions that remain to be further explored. During RL training, the system prompt plays an important role in improving the efficiency of safety enhancement. Prompts can constrain the generation format of the model, making the effect of RL rewards more explicit on model behavior. From the perspective of the essence of RL, it fixes high-quality explorations within a certain degree of freedom and maintains this tendency through iterative training. If the model’s output is not constrained, the model may spend extra “effort” on format-divergent “exploration”, resulting in

Metric	Dataset	Original	TP	Anti-TP
ACC	Letter	13.33	14.00	11.33
	CSQA	52.83	54.05	43.82
	GSM8K	49.28	52.92	46.32
	StrategyQA	53.71	54.93	51.62
ASRc	Letter	20.00	12.00	17.33
	CSQA	20.72	18.84	27.27
	GSM8K	22.21	20.17	27.07
	StrategyQA	46.03	44.98	48.21

Table 3: Performance comparison across different conditions. We use ACC and ASRc as indicators. The green font in the table represents positive improvement, while the red font represents negative improvement. Experiments on Anti-TP show that in most cases, even if RL is not stable, the reverse application of TP design has the potential for design attacks.

only minor improvements in safety.

Q1 and Q2: Intelligent LRMs are easier to manipulate.

Qwen3, a newer LRM, outperforms Deepseek on most reasoning datasets but shows significantly weaker robustness against BadChain attacks—failing completely on Letter. This means that LRM with higher inference performance has the potential risk of being more vulnerable to CoTA. The attack methods and permission requirements of CoTA only need user-level operations. Due to its higher reasoning ability, Qwen3 can discover more of the user’s intentions. It can dig out the relationship between triggers and redundant reasoning steps more deeply than Deepseek. This leads to the model behaving more stubbornly under the backdoor reasoning examples indicated by the attacker.

Q2: Reasoning experts are the stubborn ones. By comparing the CR and RR indicators in two tables, we find that Meta-Llama-3.1-8B-instruct, as a common LLM, is more likely to be “treated” under the TP framework. We suspect that this might be during the training stage, the training of LRM for CoT leads to an enhancement of the logic among the corpora. This will lead to LRM having a closer contextual relationship under the malicious manipulation of prompts, and thus being more susceptible to the injection of the CoT.

5.2 Q3: Effectiveness and Attack Potential

To Anti-TP models, the reward behavior will be contrary to the TP designed strategy logic. Overall, we consider the following three situations:

- **Setting the reward value to 0.** Obviously, when the test temperature is set to 0 and no reward is applied, the model’s performance does not change.
- **Setting the reward from X to $-X$.** Compared to “reversed” attacks, the model tends to “do nothing”. Although both performance and security decrease, the attack effect becomes unstable due to the influence of training temperature [Guo *et al.*, 2025]. In this case, essentially, RL rewards action A, while $-X$ corresponds to punishment. This causes positive attempts made by the model at training temperature to be eliminated.

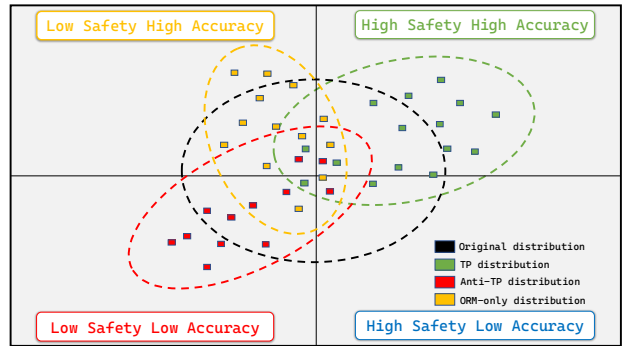


Figure 3: The distribution of the model performance under different RL Settings. In a large number of repetitive experiments, RL demonstrates its unstable characteristics. Even minor parameter changes can lead to deviations in the results. However, in terms of distribution, the three different RL Settings generally have a distinct tendency, roughly as shown in the figure.

- **Rewarding behavior from action A to $-A$.** In this case, the reward module will try to expand “malicious” exploration and retain it during the training iterations, which is more in line with the logic of attack methods. Therefore, Anti-TP, as an exploration of the attack potential of the TP defense process, adopts the third setting. Compared with the second design, it is more interpretable, has a higher attack upper bound, and more strongly proves the role of RL in backdoor security.

Our results table 3 indicate that the reverse setting of the TP framework has the potential for model attack angles. Under the setting of Anti-TP, LRMs break through the lower limit of the original performance, and such performance seems stable and powerful. This setting may be injected by the backdoor during the training process of the LRMs, completing the specific content of the tampering reward behavior. Furthermore, as an independent part from the CoTA, new attack combinations also have the value of exploration.

5.3 Case Study and Further Analysis

In the experiment table 4, we observe the output of the model during training and find that the effective responses of the model can be roughly divided into four types: For the Bad Answer: In this case, the model completes the harmful reasoning steps completely in accordance with the attacker’s will without any awareness, which is a normal performance under the BadChain attack. For the Warning Answer: The model detects that the problem contains suspicious characters and gives a warning statement containing labels without affecting the generation of the thought chain. However, the reasoning of the model is still affected by the injection of prompt words and requires manual calibration. For the Clean Answer: The model gives warnings, marks and skips redundant reasoning steps. At this point, the model realizes the correct answer, but it cannot completely prevent the output of harmful content. It requires the cooperation of an external script to achieve perfection. For the Perfect Answer: The model issues warnings and directly ignores harmful content, which is efficient and accurate. However, this situation rarely occurs in experiments